

From Asset Administration Shells to Knowledge Graphs

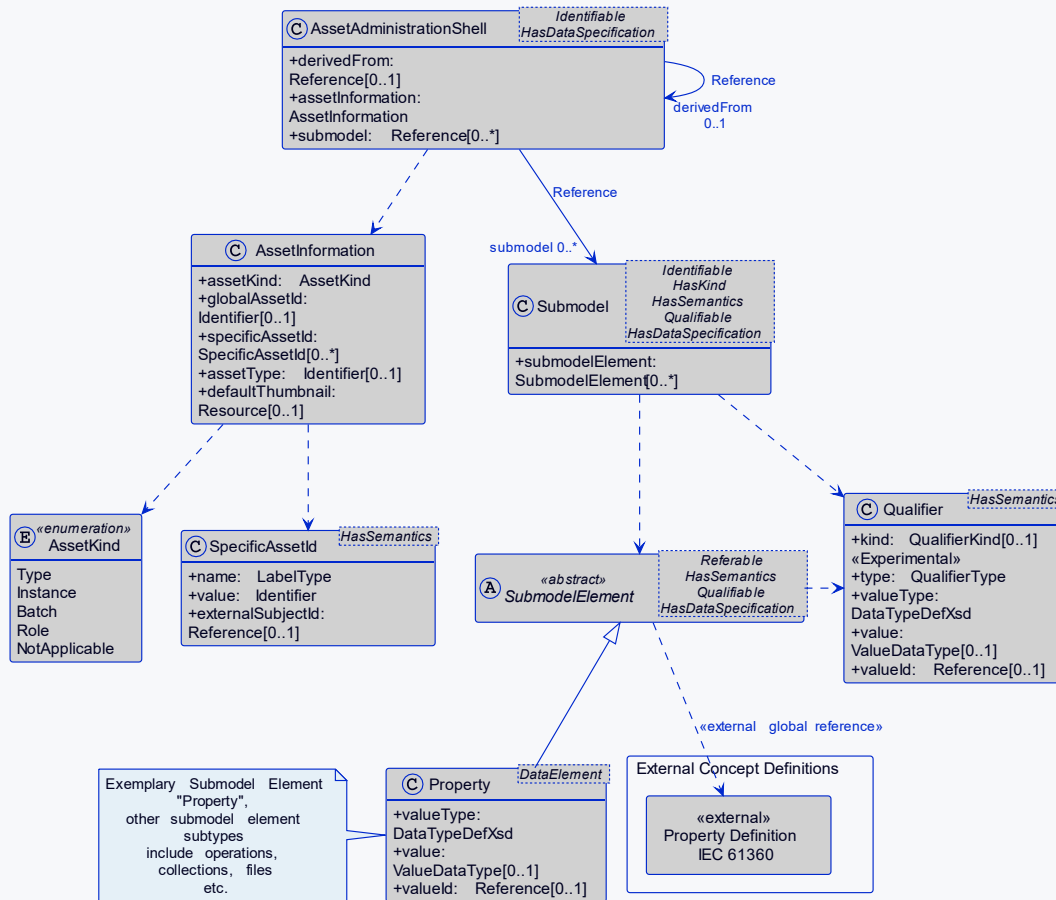
Rule-Based Semantic Lifting for Cross-Domain Reasoning in Cyber-Physical Systems

Harish Kumar Pakala,
Bianca Wiesmayr, Christian Diedrich
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**How can structured asset information
become machine-interpretable knowledge?**



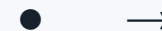
What is an Asset Administration Shell (AAS)?



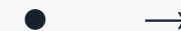
An AAS is a standardized digital representation of an asset.

- Identifies the asset
- Organizes information
- Uses **Submodels** to describe to different aspects

ASSET



AAS



SUBMODELS



Example: A **Sensor** can have Submodels for technical data, documentation, and maintenance.



AAS Submodels: A Concrete Example

ConditionMonitoring

- ⊢ AssetId = Pump_01
 - ⊢ semanticId = "aas:AssetIdentifier"
- ⊢ DeviceClass = "CentrifugalPump"
 - ⊢ semanticId = "eclass:CentrifugalPump"
- ⊢ BearingTemperature = 78.4
 - ⊢ semanticId = "qudt:Temperature"
- ⊢ VibrationVelocity = 4.8
 - ⊢ semanticId = "qudt:Velocity"

Condition Monitoring of a centrifugal pump

1. Asset information

Pump_01 identifies the physical asset.

2. Engineering information

CentrifugalPump, temperature, and vibration describe the asset's condition.

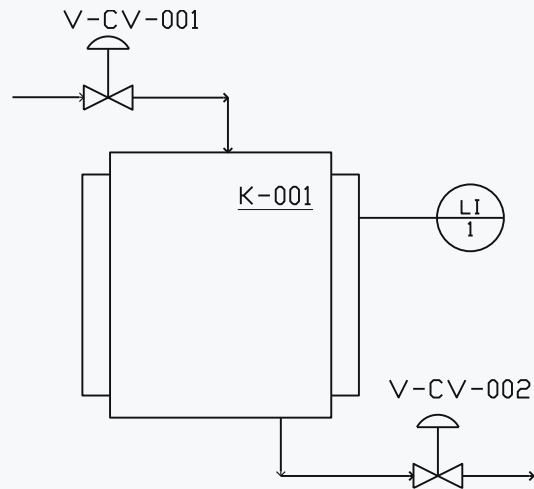
3. Semantic references

eCl@ss and QUDT provide machine-interpretable meaning.

The Submodel combines **asset data** with **semantic references** — but the relationships between **engineering concepts** are still limited.

From Engineering Models to AAS Submodels

Process Engineering



- Process equipment
- Instrumentation
- Actuating functions
- Instrumentation functions

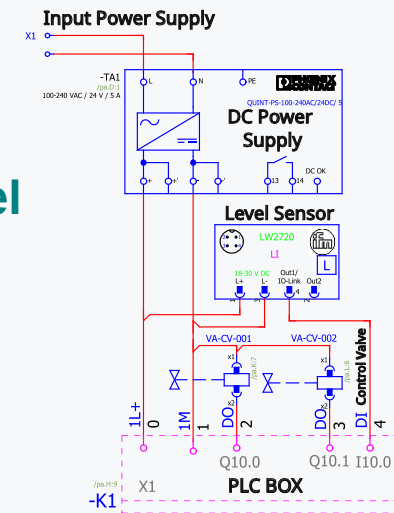
DEXPI AAS Submodel

ECAD AAS Submodel

Common AAS Metamodel

Semantic Integration

Electrical Engineering



- Devices
- Symbols
- PLC endpoints
- Wiring / connections

AAS Submodels provide a **common structure** for representing **heterogeneous engineering information**.

From Structured Information to Engineering Knowledge

What AAS gives us

- Structured engineering information
- Semantic references
- Asset identifiers
- Explicit AAS relationships

AAS Submodels



**How can this information
become machine-
interpretable
engineering knowledge?**



Reasoning & Querying

What we want to do

- Understand relationships between concepts
- Query engineering information
- Combine information from different Submodels
- Derive new information through reasoning

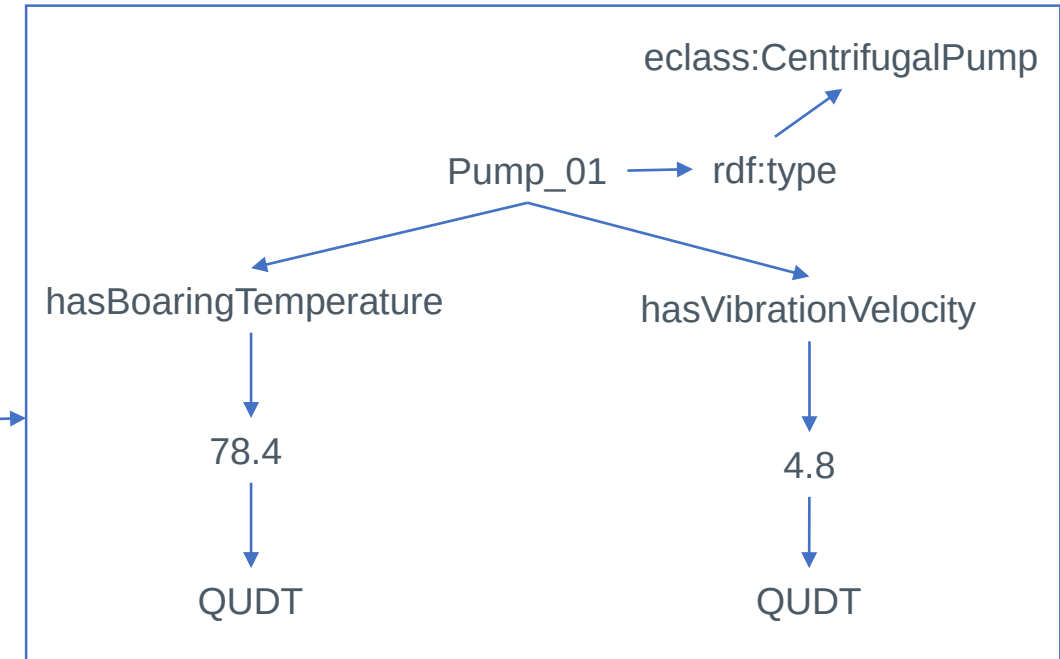
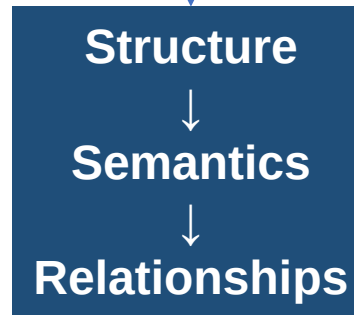
AAS organizes engineering information — but we need a representation that makes its semantics and relationships explicitly usable for querying and reasoning.

Semantic Lifting: From AAS to Knowledge Graphs

ConditionMonitoring

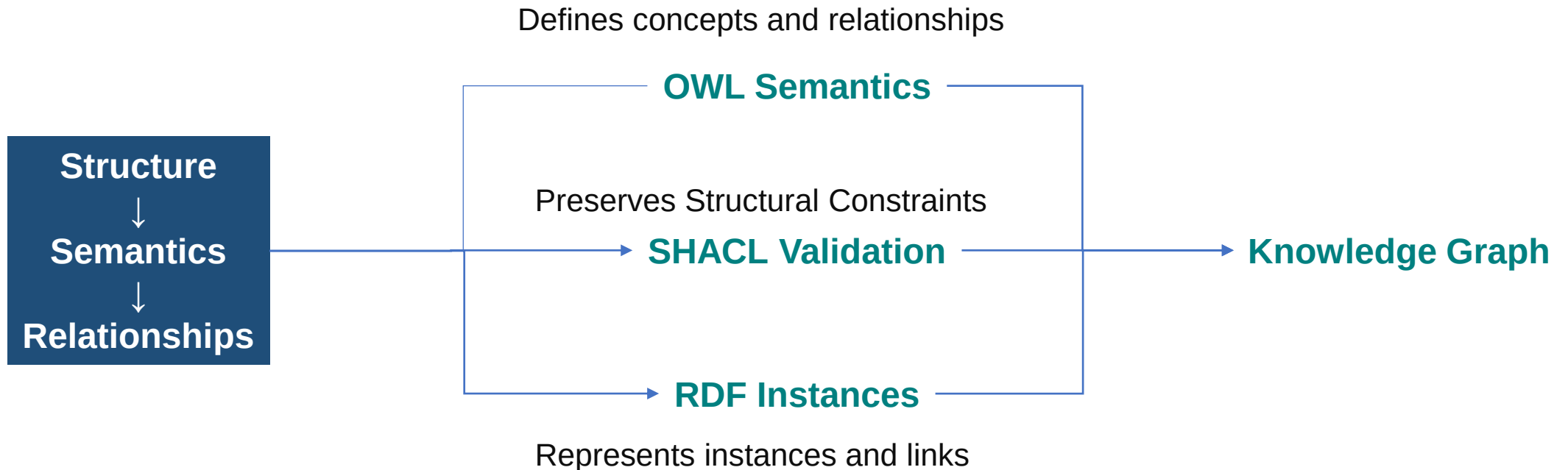
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Semantic Lifting



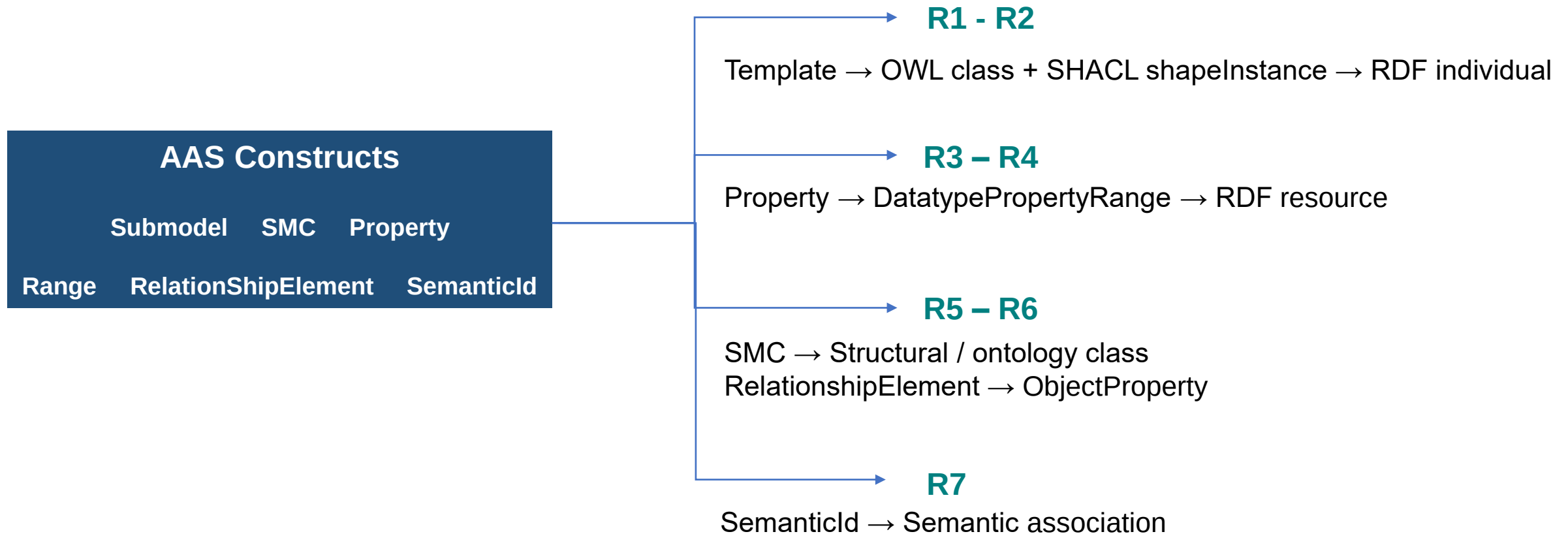
Semantic lifting turns structured AAS information into an explicit network of concepts and relationships.

How does Semantic Lifting Work ?



Semantic lifting turns structured AAS information into an explicit network of concepts and relationships.

Semantic Lifting Rules R1 – R7



Semantic lifting turns structured AAS information into an explicit network of concepts and relationships.